

CS2AI Artificial Intelligence Technical Report

Project Title: Predicting Income Level Using Census Data

Dataset: UCI Adult Dataset

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1. Abstract

In this project, we built a machine learning pipeline to predict income using the UCI Adult Census dataset. This goal involved determining whether individuals earned over \$50,000 annually based on age, education, occupation, marital status and hours worked. This was to address a problem of government attempting to target wealth inequality policies more accurately. Our data preparation relied on one-hot encoding, standard scaling and stratified splitting. Comparing two models came with trade-offs. A Random Forest Classifier achieved 85.3% overall accuracy. A class-weighted LASSO Logistic Regression model offered an interpretable baseline with 81.2% balanced accuracy and 83% minority class recall. The logistic regression provided stronger equity across income groups. These metrics highlight an accuracy problem in imbalanced classification tasks. Furthermore, working with sensitive demographic attributes requires evaluation of bias and fairness.

2. Background and Problems to be Addressed

The Adult Census Income dataset contains demographic and employment related attributes for individuals, with the goal of predicting whether a person earns more than \$50,000s per year. This prediction problem is relevant in domains such as labour market analysis, socio-economic research, and policy evaluation, as it provides insight into the factors associated with income equality. As such the problem this project addresses is a scenario where the US Department of Labour are attempting to target financial support and wealth inequality relief programmes more accurately. The model aims to give the government a reliable way to predict which individuals are likely to below a given threshold (\$50,000). The dataset includes both numerical and categorical features, such as age, education level, occupation, marital status, race, gender and hours worked per week. These variables capture a broad range of socio-economic characteristics, making the dataset suitable for exploring relationships between demographic factors and income levels. The problem addressed in this project is to build machine models capable of accurately predicting ($>50K$ or $\leq 50K$), so that the government can target policies that tackle wealth inequality. This involves performing exploratory data analysis (EDA), applying appropriate preprocessing techniques, and evaluating multiple models to determine which approach performs best. The task also requires consideration of fairness, bias, and ethical implications, as income predictions intersects with sensitive demographic attributes.

3. Ethical and Social Aspects

The Adult Census dataset contains sensitive demographic attributes including gender, race, and marital status as these features tie directly to systemic inequality. We face three core ethical problems. First, the data reflects real-world societal biases as we risk reproducing these patterns when training algorithms. This causes harm in high-stakes applications including hiring, credit scoring, or resource allocation. Second, we must justify including sensitive attributes as these inputs improve predictive performance while risking discriminatory outcomes. We balance accuracy with fairness and accountability. Third, the anonymized dataset implicates GDPR data protection principles. We must handle data

responsibly. We analysed sensitive features with these principles in mind throughout our project and interpreted model performance while monitoring bias and social impact.

4, Analysis Design

The analysis follows a structured workflow to ensure clarity, reproducibility, and fairness in the modelling process.

EDA is used to understand the distribution of key variables, identify patterns, detect class imbalance, and highlight potential sources of bias. Visualisations such as histograms, count plots, and correlation heatmaps help reveal relationship between demographic attributes and income. The dataset contains mixed datatypes and missing values represented as “?”.

Preprocessing includes cleaning, encoding categorical variables, scaling numerical features, and splitting the data into training and testing sets. A unified preprocessing pipeline is built to ensure consistency across all models. Multiple machine learning models will be trained using the preprocessed data. Each model will be evaluated using appropriate metrics such as accuracy, precision, recall and F1-Score. Stratified Sampling is also used to preserve class balance. Results from different models will be compared to determine which approach performs best. Interpretation focuses not only performance metrics, but also on fairness considerations and the influence of key features. This structured design ensures that the analysis is systematic, transparent, and aligned with the best practices in machine learning. The full analysis pipeline flow chart is shown in the appendix in Figure 8.

5. Exploratory Data Analysis (EDA)

For the following plot figures, see their corresponding images in the EDA appendix.

The income distribution (**Figure 1**) is highly imbalanced, with the majority earning $\leq 50K$, indicating that accuracy alone will not be a reliable evaluation metric and that additional metrics such as precision, recall and F1-score will be necessary, alongside stratified sampling during splitting. The age distribution (**Figure 2**) is concentrated between 20 and 50 years old and is right-skewed, suggesting a potential non-linear relationship between age and income.

Figure 3 shows a clear upward trend between education level and likelihood of earning $>50K$, with higher education categories such as Bachelors, Masters and Doctorate containing noticeably more $>50K$ earners, confirming education as a strong predictor. The gender distribution (**Figure 4**) reveals a significant disparity, with males having a much higher proportion of $>50K$ earners, highlighting potential gender-based income inequality that models may inadvertently reinforce. Similarly, **Figure 5** shows that White individuals have a noticeably higher proportion of $>50K$ earners compared to all other racial groups, suggesting racial disparities that should be considered when interpreting outputs. Weekly working hours (**Figure 6**) peak strongly at 40 hours with a long tail toward 50–60+, meaning this feature may contribute meaningfully to model performance. Finally, the correlation matrix (**Figure 7**) shows generally weak correlations among numerical features, with the strongest being `education_num` against `hours_per_week` (0.15) and `capital_gain` (0.12), suggesting that non-linear models may capture relationships more effectively.

6. Data Pre-processing, Feature Engineering and Feature Selection

The Adult Census Dataset requires several preprocessing steps before it can be used for machine learning. These steps ensure data quality, consistency and suitability for numerical algorithms. The dataset uses “?” to represent missing entries in several categorical columns. These values were replaced with “None”/”NaN” and removed using *dropna()*. Given the dataset’s size, removing incomplete rows does not significantly reduce data availability and avoids introducing noise through imputation. The dataset contains multiple categorical features such as workclass, education, occupation, and marital status. These were encoded using *OneHotEncoder* which converts each category into a binary indicator. This approach avoids imposing ordinal relationships where none exist and ensures compatibility with linear and non linear models. The encoder was configured with *handle_unknown* = “ignore” to prevent errors when encountering unseen categories during testing. Numerical features (e.g, age, hours_per_week_, capital_gain) vary widely in scale. *StandardScaler* was applied to normalise these features to zero mean and unit variance. Scaling improves model stability and performance, particularly for algorithms sensitive to features magnitude, such as logistic regression and SVM (support vector machine). The dataset was split into training and testing sets using an 80/20 ratio with *stratify* = *y*. Stratification preserves the original class distributions, which is important because the dataset is imbalanced. A unified preprocessing pipeline was built using *ColumnTransformer* and saved as *preprocessor.pkl*. This ensures that all models use identical preprocessing steps, prevents data leakage, and improves reproducibility. The pipelines can be loaded and applied directly to the train/test splits of the models.

7. Machine Learning Model 1: Logistic Regression

7.1 Summary of the Approach

Logistic Regression is a classification technique that calculates the probability that a given record belongs to a specific category. The algorithm combines an profile attributes such as (e.g., age, education) to output a likelihood percentage of the individual that earns above 50k. If the probability scores above the 50% threshold, the model classifies the person as a high earner. Logistic Regression was chosen for two reasons. Firstly, the algorithm guarantees full transparency as the math assigns an explicit weight to every feature. This direct scoring makes predictions completely explainable for socio-economic analysis. Secondly, the model serves as a practical baseline with one-hot encoding expanding feature space to over 100 columns. Logistic regression tests whether a linear approach solves the problem before one spends compute time on complex alternatives.

7.2 Model Training and Evaluation Methodology

To handle new columns efficiently, the pipeline utilizes the **SAGA** optimization algorithm. It works by reviewing the data and making tiny corrections to feature weights until the model's accuracy stabilizes. To ensure the training process finished efficiently without sacrificing quality, the algorithm was set to stop once improvements became negligible, capped at a maximum limit of 2,000 review cycles. The census data is heavily imbalanced, with roughly 76% of records in the lower income bracket. Left unadjusted, models bias toward the majority group to inflate overall accuracy. To correct this, the model utilized balanced class weighting, which applies a heavier penalty for misclassifying the minority class. This forces the algorithm to prioritize identifying high earners.

7.3 Results and Discussion

Table 7.3: Performance Comparison of Logistic Regression Variations [\(see in appendix\)](#)

The metrics highlight why basic accuracy is misleading on unbalanced data. The unweighted models scored over 84.7% overall accuracy primarily by defaulting predictions to the "low income" majority, thereby failing to catch actual high earners. Activating class weights (Models 4 and 5) dropped overall accuracy slightly to 80.6%. However, **Balanced Accuracy** jumped from 76.7% to 81.3%, proving the model successfully restored sensitivity to the wealthy minority. The LASSO configuration was highly effective at managing the 100+ One Hot Encoded columns. By applying strict L1 selection penalties during training, LASSO purged 19 redundant feature columns, reducing data clutter without sacrificing baseline reliability. Inspecting the surviving features confirmed real-world economic logic: heavy capital gains, advanced education, and executive/managerial roles were identified as the absolute strongest indicators of earning over \$50K.

8. Machine learning model 2: Random Forrest Classifier

8.1 Summary of Approach

This section implements and evaluates a Random Forest Classifier and see its suitability in the problem of predicting whether an individual earns over \$50,000.

While a single decision tree is highly interpretable, they are known to overfit training data severely, and exhibit instability when exposed to new information. One method of overcoming these limitations is a Random Forest Classifier, which was selected as the second model. The Random Forest algorithm is an ensemble method that tries to combat overfitting by training multiple decision trees on bootstrapped samples, meaning each tree is trained on a slightly different random subset of the data (Grus, 2019, pp. 290-291). Important to note that each subset is drawn with replacement, in other words they are placed back into the group of possible selections on each subsequent draw (probability of selection never changes).

Simply bootstrapping the trees can lead to highly correlated models, in the case of dominant predictors (as every tree in the ensemble will realize the dominant predictor is the best indicator and use it for the first split at the top of the tree). To solve this Random Forests, introduce a second layer of randomness. The model is only allowed to search for the optimal split from a randomly selected subset of features at each node, this restricts the splitting process and forces diversity (decorrelation) (Russel and Norvig, 2022, pp. 714-716).

By combining these two aforementioned techniques (bootstrapping the data and randomizing feature selection) and taking a majority vote across the ensemble, the Random Forest massively reduces model variance and single tree instability. The ensemble approach leads to better predictive accuracy on unseen census data.

8.2 Model Training and Evaluation

The modelling process utilised the census income dataset (adult.data) The data was split between 80% training set and 20% testing set, utilising stratified sampling to preserve the imbalanced distribution of the target income across both sets. Preprocessing was applied to data prior to training. The Random Forest Classifier was then instantiated and trained on this transformed data using the Shannon entropy criterion. This metric is a mathematical way to

evaluate the quality of potential splits. To evaluate the model, predictions were generated on the testing set (unseen by the model) and assessed using Overall Accuracy, Precision, Recall, F1-Score and Confusion Matrices. Mean Decrease Impurity (MDI) was used to assess feature importance on the training data, while Permutation Feature Importance (PFI) was used to measure the change in prediction error when shuffling features, this was done over 50 repeats.

8.3 Results and Discussion

The Random Forest model achieved an overall accuracy of 85.3% on the testing data. Analysing the performance on the two income classes reveals distinct differences in performance. Less than or equal to 50K: The model performed exceptionally well on the majority class, achieving a precision of 88.4% and a recall of 92.6%. Greater than 50K: Performance was lower for the minority class, achieving a precision of 73.9% and a recall of 63.2%

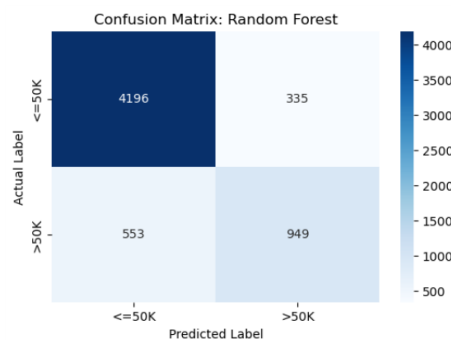


Figure 1: Confusion Matrix

The confusion matrix visually illustrates the performance imbalance. The ensemble model correctly identified 949 people who earned over 50K and controlled false positives to 335 instances; however, it struggled with false negatives, falsely predicting 553 were low earners. The model failed to identify a significant number of high earners.

MDI and PFI were used to understand the random forest's predictions. MDI evaluates importance during the training phase and can be sensitive to features with high cardinality (features like final weight which represents the number of people represented by a row and has absolutely no correlation to income). PFI was calculated over 50 repeats on the unseen test data to provide an estimation of feature importance. When a feature's relationship with a target is broken, a drop in a predictive accuracy can be observed.

The PFI analysis confirmed that capital gain and age caused the most significant decrease in accuracy when shuffled as they are strongest true predictors of income (see appendix A for full Permutation importance charts). These were followed in importance by capital loss, marital status, hours worked per week and education level (education_num). Features at the bottom of the chart showed very little permutation importance, indicating they were redundant as standalone predictors and the primary financial and age-related features are significantly more important, which aligns with real world.

9. Results Interpretation Comparing Across the Models Built

Using the same 80/20 stratified train-test split and the same preprocessing approach, the Logistic Regression and Random Forest models were evaluated, which made the comparison fair, as after consistent handling of missing values, categorical encoding and numerical scaling, both models were tested on the same unseen data. We didn't use accuracy on its own since the Adult Census Income dataset is imbalanced with the less than or equal to 50K class appearing more than the more than 50K class. To see performance across both income classes a lot more clearly we also consider balanced accuracy, precision, recall, F1-score, MCC and confusion matrix.

Overall accuracy can be misleading for this dataset, and this is shown in the results. The standard logistic regression model has 84.7% accuracy, however its balanced accuracy was only 76.7% meant the model performed better on the majority less than or equal to 50K class than on the minority more than 50K class. The Weighted Ridge logistic regression model brought down overall accuracy to 80.6% but increased balanced accuracy to 81.3% while still maintaining MCC of 0.57 and this shows that instead of mainly favoring the majority class, class weighting helps the model to treat both income groups a lot more evenly.

The random forest classifier has the highest accuracy overall at 85.3% which makes it the strongest model for general predictive performance and this is due to the fact that Random Forests are able to learn more complex and nonlinear relationships between features than Logistic Regression can but its results still had an issue for the minority more than 50K class. The model has a precision of 0.74, recall of 0.63 and F1-score of 0.68 for this class (Appendix A, Figure A2), meaning that when it predicted more than 50K, it was for most times correct, but it still missed many that actually earned more than 50K and this is supported by the confusion matrix as 949 high-income records were classified correctly while 553 were classified wrong as less than or equal to 50K. (Appendix A, Figure A1).

A limitation of this evaluation would be that although stratified sampling and class weighting do help in issues with imbalance, they don't prove fairness across groups like race and sex, so to check whether errors are distributed evenly or not in the future, comparison of performance should be carried out separately across these groups and for stronger evidence of stability of the model and the chance of overfitting cross validation or training vs testing score comparisons would be useful to carry out.

Overall, it depends on the aim. If prediction accuracy is the main aim, its better to use Random Forest but if the main aim is interpretability or transparency and balanced treatment of both income classes, weighted ridge logistic regression would be better.

Table 9: Cross Model Performance Evaluation [\(see in appendix\)](#)

These comparative results demonstrate the accuracy paradox. The Random Forest reaches 85.3% overall accuracy by mapping the majority class well and achieving 92.6% majority recall. Without class imbalance penalties, the model produces 553 false negatives and a minority recall of 63.2%. The Logistic Regression applies an asymmetric penalty through balanced class weighting which trades raw accuracy for an 81.2% balanced accuracy and increases minority recall to 83.0%. Both models identify the same predictors. LASSO drops

19 coefficients to zero. Capital Gain at +2.50 and Civil Marriage at +1.56 act as the top positive drivers in the logistic model. Permutation feature importance shows that Capital Gain and Age cause the largest accuracy drops in the Random Forest. Marital Status and Education Level follow closely. The Random Forest captures nonlinear interactions while remaining unclear. Logistic Regression translates coefficients directly into odds to support policy decisions requiring transparency.

10. Conclusion, Recommendations and Future Work

10.1 Conclusion

Overall, the project's objective of developing a machine learning pipeline capable of predicting whether an individual earns above or below \$50,000 based on census data, was successfully met. This objective was met through implementing and comparing a class weighted LASSO Logistic Regression model against a Random Forest Classifier.

Comparing the two models revealed a clear example of the accuracy paradox. The Random Forest Model achieved the highest overall accuracy at 85.3% due to its flexible nonlinear mapping, which makes it superior for pure classification purposes. However, because it lacked explicit class imbalance penalties, it struggled with the minority class and achieved a poor recall of 63.2%. Conversely the Logistic Regression model utilised balanced class weights, trading accuracy for distributional balance; it achieved a 81.2% balanced accuracy and a significantly higher recall of 83% on the minority class. Both models converged on accepted socio-economic realities. Feature interpretability analysis, using L1 regularisation pruning in logistic regression and PFI in the random forest confirmed capital gain, age and marital status as the strongest predictors of income. Ultimately, the optimal model depends on the use case objective; Random forests excels in predictive performance while logistic regression provides minority class equity, for targeted policy intervention that is particularly important. However, in the realm of policy creation, the PFI ranking may actually be a more useful indicator than the model, if the Department of Labour wanted to address inequality, discovering the root differences between earner groups, might be more useful than a relatively low accuracy binary predictor.

10.2 Recommendations and Future Work

Based on ethics and the model analysis, model selection depends on the use case. Select the Random Forest to prioritize raw accuracy and deploy the class-weighted Logistic Regression to ensure transparency and equal treatment of income groups. It's important to mandate human oversight for real-world decisions about individuals. Future work could require k-fold cross-validation to confirm stability across data splits. We also need to tune hyperparameters by varying tree depth, estimator counts in the Random Forest and conduct sub-group fairness analysis across sex and race to verify equitable misclassification rates. Testing ensemble boosting methods like XGBoost offers better minority class recall.

11. References

Grus, J. (2019) *Data science from scratch: first principles with Python*. 2nd edn. Sebastopol, CA: O'Reilly Media.

Russell, S. and Norvig, P. (2022) *Artificial intelligence: a modern approach*. Global edn. Harlow: Pearson Education.

12. Group reflection

Our project required coordination across five distinct areas. Dividing the work gave each member clear ownership while presenting challenges in ensuring consistency across the final report.

Hassan focused on the data selection, exploratory data analysis and preprocessing. He sourced the UCI Adult Census dataset and justified the dataset for a classification task, made visualisations covering income distribution, age, education, gender, race and hours worked and then built the preprocessing pipeline using ColumnTransformer and StandardScaler and both models relied on this pipeline. Through this, Hassan learned how preprocessing decisions directly affects model fairness with stratified splitting and one-hot encoding configuration altering fairness and accuracy.

Zack and Thomas took focused on the modelling sections. Zack focused on the Logistic Regression implementation whilst Thomas focused on the Random Forest Classifier. Zack looked at five configurations of logistic regression to demonstrate the accuracy problem, implemented class weighting and LASSO regularisation to improve minority class recall. Thomas implemented the Random Forest with Shannon entropy splitting, did MDI and Permutation Feature Importance analysis across 50 repeats to produce reliable importance estimates. Early in the project, Zack and Thomas worked from different pipeline versions, producing inconsistent results so we used Hassan's pipeline to deal with these inconsistencies. Through this, Zack and Thomas developed an understanding of why ensemble methods do better than single models on raw accuracy while needing extra attention on class imbalance.

Amanze managed the ethics section, abstract, conclusion and overall report quality/consistency. He looked at the ethical implications of using sensitive demographic attributes, analysing race, gender and marital status in the context of real-world decision-making and GDPR. He did the final structure review and checked the consistency of the report, writing the abstract and conclusion alongside this and then merged everyone's work into a single document. Writing the ethics section required him to review the results thoroughly, such that he now realises how high balanced accuracy fails to guarantee fairness across demographic subgroups.

Barakah handled evaluation by designing the framework for comparing models and writing the cross-model analysis. They evaluated balanced accuracy, MCC, precision, recall and F1-score then produced Table 9.2 summarizing performance across models. From this, Barakah learned the practical difference between overall accuracy and balanced accuracy in imbalanced datasets and that the choice of evaluation metric must equal the real-world cost of misclassification.

Throughout the completion of this report, our group faced many challenges. Some members completed some sections close to the deadline, leaving little time for joint review but scheduling a mid-project check in allows all members to review sections together. These check ins allowed us to catch inconsistencies early, for example, the team underestimated how ethical analysis effects modelling decisions and therefore we had to integrate ethical discussions early in future projects. This project gave us practical experience with the full machine learning pipeline from exploratory data analysis to evaluation, with the model selection depending entirely on the context of model deployment.

13. Appendices

Exploratory Data Analysis (EDA) Appendix

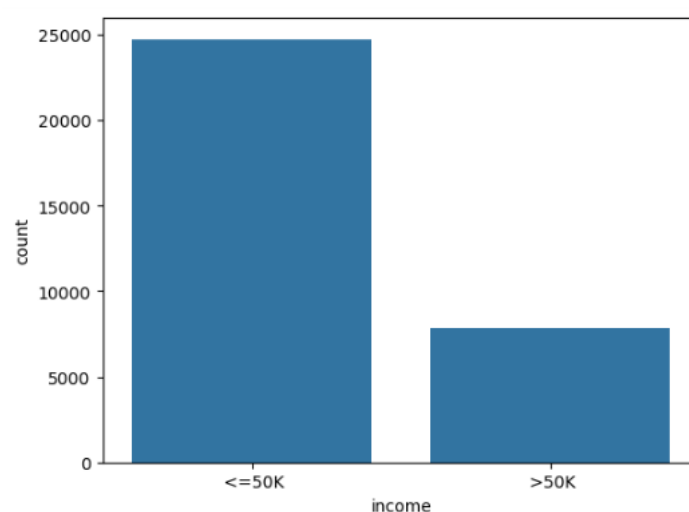


Figure 1: Income Distribution

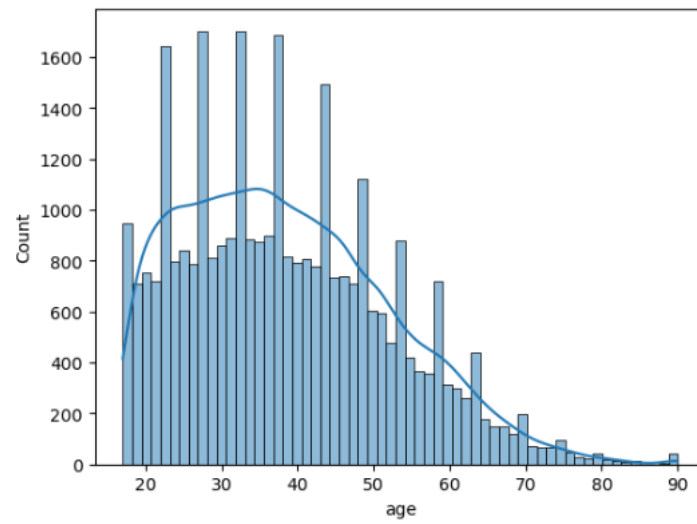


Figure 2: Age Distribution

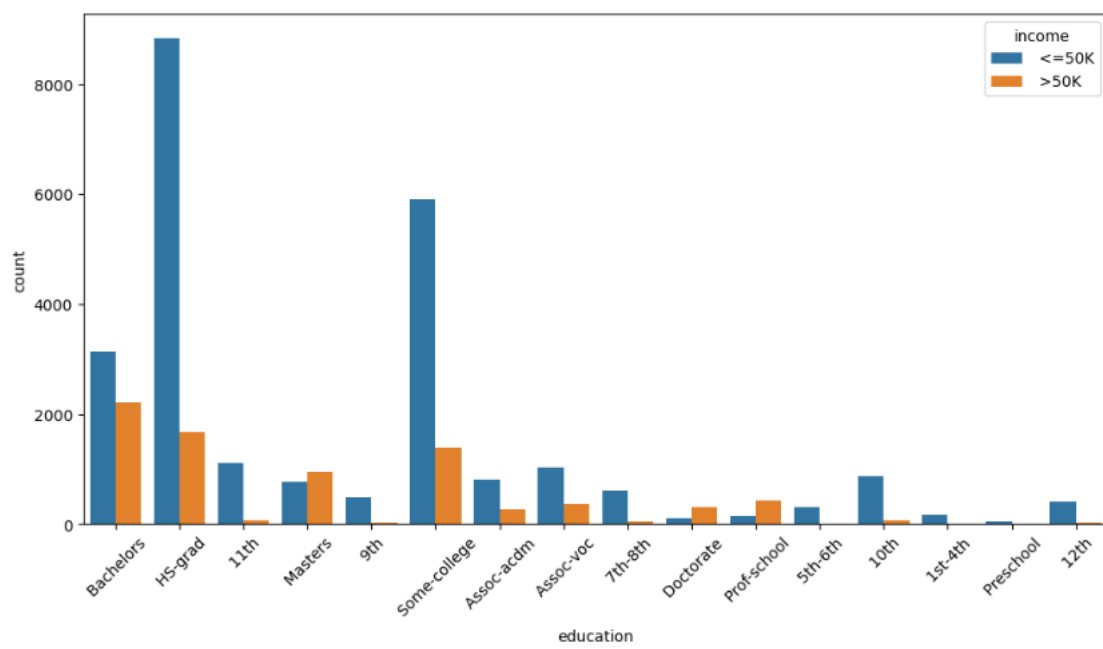


Figure 3: Education vs Income

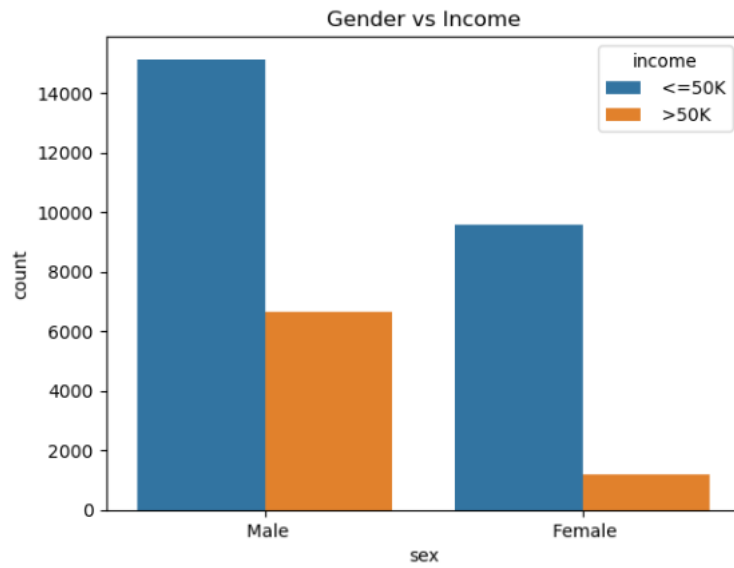


Figure 4: Gender vs Income

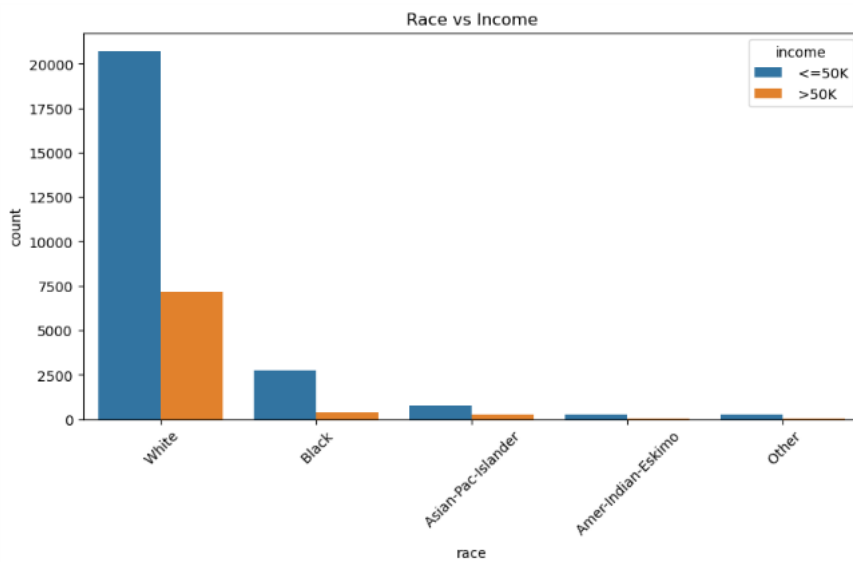


Figure 5: Race vs Income

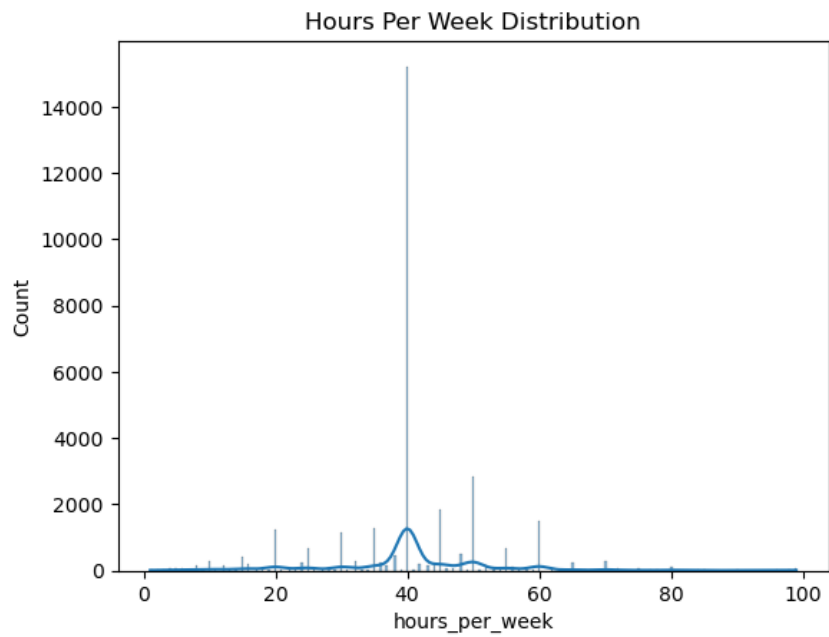


Figure 6: Hours Per Week Distribution

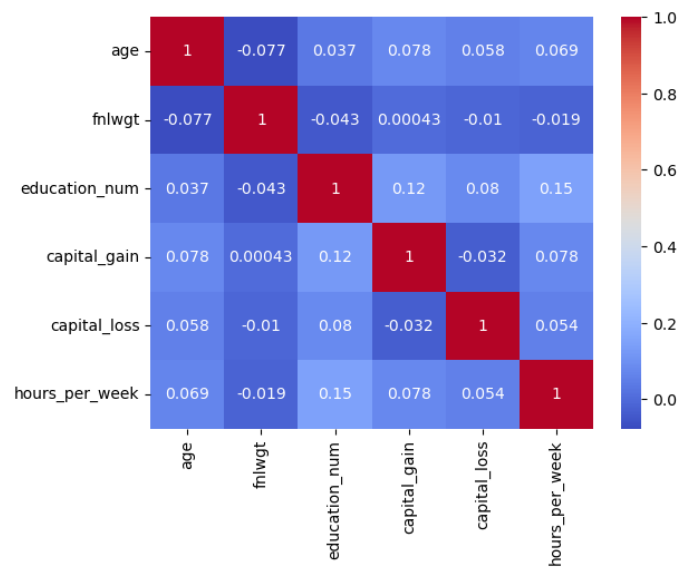


Figure 7: Correlation Heatmap

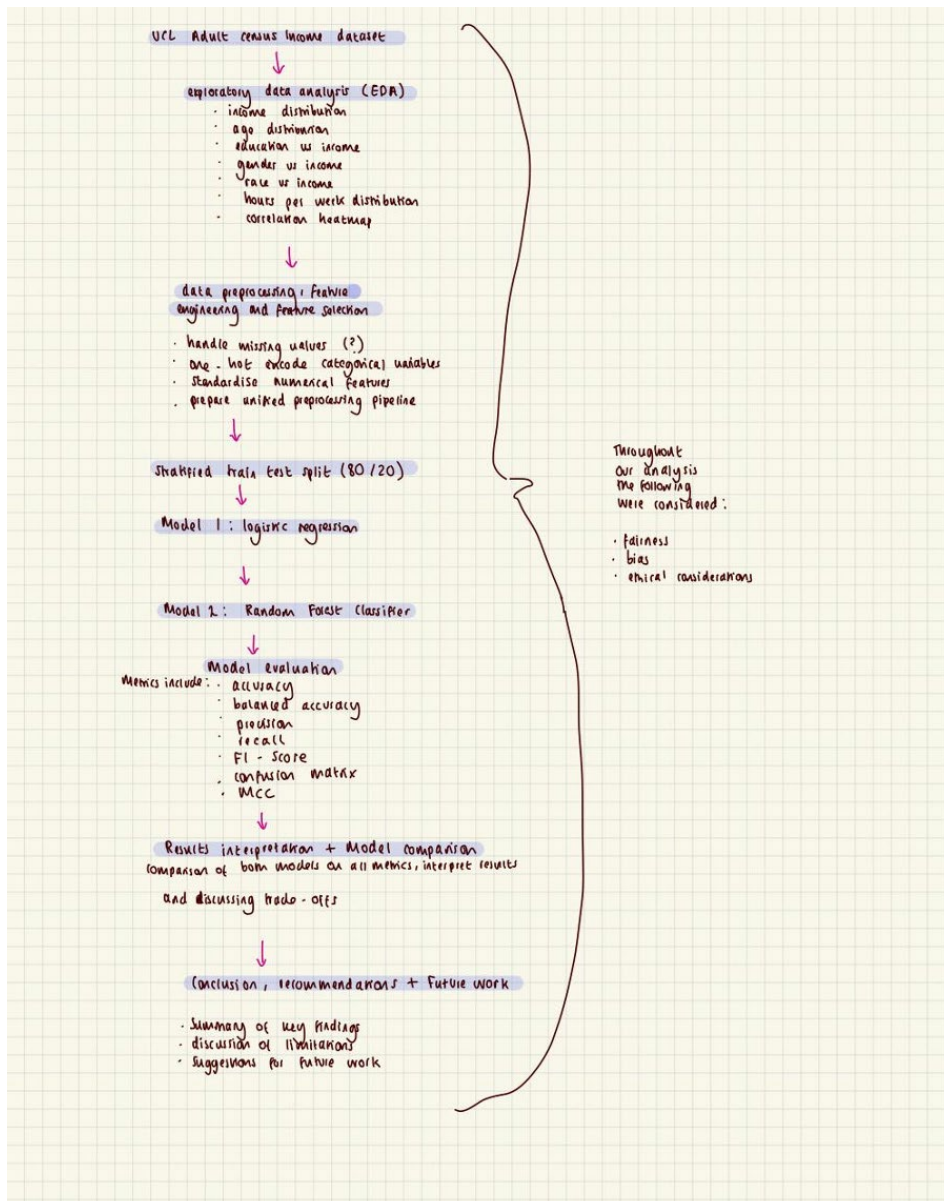


Figure 8: Analysis Design Flow Diagram

Table 7.3: Performance Comparison of Logistic Regression Variations

Model Configuration	Primary Goal	Feature Selection Strategy	Imbalance Correction	Overall Accuracy	Balanced Accuracy	Reliability Score (MCC)
1. Standard Baseline	Default Learning	None	None	84.7%	76.7%	0.57

2. Ridge Regularization	Weight Stabilization	Gentle Shrinkage	None	84.6%	76.3%	0.57
3. LASSO Regularization	Purge Clutter	Aggressive Pruning	None	84.6%	76.5%	0.57
4. Weighted Ridge	Fair Detection	Gentle Shrinkage	Active Weights	80.6%	81.3%	0.57
5. Weighted LASSO	Maximum Balance	Aggressive Pruning	Active Weights	80.6%	81.2%	0.56

Appendix A

	Precision	Recall	F1-score	Support
<=50K	0.884	0.926	0.904	4531
>50K	0.739	0.632	0.681	1502
Accuracy	0.853			
Macro Average	0.811	0.779	0.793	6033
Weighted Average	0.848	0.853	0.849	6033

Figure 3: Random Forest Classification Report

Figure 4: PFI

Table 9: Cross Model Performance Evaluation

Evaluation Metric	Baseline Logistic Regression (Unweighted)	Final Logistic Regression (Class-Weighted LASSO)	Final Random Forest Classifier (Unweighted Ensemble)

Primary Optimization	Global Log-Loss	Penalized Log-Loss (L1 Sparsity)	Shannon Entropy (MDI/PFI Splits)
Class Weighting	None	Balanced	None
Overall Accuracy	84.6%	80.6%	85.3%
Majority Recall ($\leq 50K$)	$\sim 93.0\%$	$\sim 79.0\%$	92.6%
Minority Recall ($> 50K$)	$\sim 58.0\%$	$\sim 83.0\%$	63.2%
Balanced Accuracy	76.5%	81.2%	77.9%
Reliability Score (MCC)	0.57	0.56	$\sim 0.55^*$